



1-TOPS team members

This ensures that even as senior students graduate, there remains a robust core of talent to complete the Tape-Out.

Prospective participants should possess a foundational understanding of digital electronics and computer architecture. Most importantly, we are looking for students with solid enthusiasm, strong motivation, and a keen passion to learn all facets of processor and chip design.

What role will the VLSI Society of India play in guiding the students to design a RISC-V core and SoC?

The VLSI Society of India and C-DAC are partnering to facilitate 1-TOPS teams in developing their RISC-V cores and SoCs. A cornerstone of this programme is the vital support provided by 1-TOPS coaches - industry volunteers, each bringing over a decade of experience in advanced chip design.

Each 1-TOPS team will benefit from a dedicated 1-TOPS coach who commits 1-2 hours every weekend to guide and facilitate their progress. It is truly inspiring

that over 30 highly experienced chip designers from the industry have already volunteered their expertise for this crucial role.

Further elevating the programme, we are honoured to have global RISC-V leaders as our Chief Mentors: Prof. Kamakoti of IIT Madras (renowned for the Shakti Platform) and Prof. Luca Benini of ETH Zurich (known for the Pulp Platform). Their involvement will not only motivate and inspire students but also provide invaluable access to the broader global RISC-V community for support and collaboration.

Which process node are you targeting for the 1-TOPS programme, and how will students access the PDK for the target node?

The 1-TOPS programme targets the 65nm or advanced process technology node, leveraging partnerships with leading global fabs such as TSMC, GlobalFoundries, Tower, and the Indian fab of TEPL. The precise node for each project will be collaboratively decided by the

1-TOPS coaches and their respective teams.

The VLSI Society will play a crucial role in supporting teams by facilitating the acquisition of process design kits (PDKs) from these foundries. Additionally, they will provide essential interface IPs (including AXI, AHB, APB, SPI, I2C, UART, and others) critical to the successful development of RISC-V SoCs. We applaud the Ministry of Electronics and IT (MeitY)'s C2S initiative to provide commercial EDA tools from leading vendors to more than 300 institutions, which has helped the VLSI Society of India to conceive the 1-TOPS programme.

MPW manufacturing for 65nm nodes or advanced nodes is expensive. How are you planning to manage the cost of MPW manufacturing?

Yes, 65nm or advanced-node manufacturing will involve certain expenses that will depend on die size. We expect that the typical MPW cost for 1-TOPS SoC chips will be approximately 800,000 to 1,000,000 rupees. The VLSI Society is delighted to announce a significant financial commitment covering 50% of each team's multi-project wafer (MPW) cost, up to 500,000 rupees. Furthermore, the Society will actively support participating institutions in securing the remaining 50% of funding from industry or government, thereby ensuring comprehensive resources for a successful Tape-Out.

Do you have any message for the aspiring students?

In response to the pressing demand for seasoned chip design engineers, the 1-TOPS programme is meticulously designed to develop state-of-the-art expertise among students, thereby unlocking remarkable career pathways. This initiative is focused not only on nurturing industry-ready talent but also on solidifying India's standing as the global epicentre of RISC-V innovation. **EFY**